



JESÚS DANIEL CABALLERO COLINA

INDUSTRIAL AI ENGINEER

Edge ML Systems | Predictive Maintenance | Oil & Gas

CONTACT:

PHONE: (+57) 3005051195

EMAIL: jesuscaballero0@gmail.com

LINKEDIN: <https://www.linkedin.com/in/jdcaballeroco/>

PROFESSIONAL SUMMARY:

Industrial AI Engineer with experience designing and deploying machine learning systems for predictive maintenance and industrial asset monitoring in Oil & Gas environments.

Currently leading the technical design and architecture of AI-driven solutions within the Proctek IA ecosystem, building end-to-end systems that integrate machine learning, backend services (FastAPI), and edge deployment (HCC2, Proton, PLCNext) using Docker/Podman in OT environments.

Developed production-grade applications for predictive monitoring of pumps and compressors, including anomaly detection, time-series forecasting, asset health metrics, and real-time industrial dashboards. Involved in technical decision-making, mentoring junior engineers, and delivering technical demonstrations to clients.

I am interested in building robust industrial AI systems designed for edge deployment and continuous operation in constrained environments.

PROFESSIONAL EXPERIENCE:

	<u>Proctek S.A. – Control Engineer (Industrial AI)</u>
	- Lead the technical design and architecture of AI systems for predictive maintenance and monitoring of industrial assets in production environments. - Develop end-to-end applications integrating backend services (FastAPI), machine learning models, and industrial visualization, enabling continuous asset monitoring. - Design predictive monitoring systems for pumps and compressors, including Anomaly detection, Time-series forecasting, Asset health and efficiency metrics - Deploy solutions on industrial edge devices (HCC2, Proton, PLCNext) using Docker/Podman on ARM architectures, operating under limited connectivity constraints. - Contribute to the development of AI agents for industrial data interpretation. - Define key technical decisions and mentor junior engineers. - Lead technical demos and solution validation with industrial clients. - Build industrial-grade products designed for continuous operation in OT environments.
Marzo 2024	
Actualidad	
	<u>Quynza Technologies – Mechatronics Engineer</u>
Marzo 2022	- Developed mechatronic solutions integrating hardware (PCBs, microcontrollers) and software for industrial systems.
Enero 2024	- Programmed embedded systems (Arduino, ESP32) for control and data acquisition. - Designed and validated HMI interfaces for physical system interaction.

EDUCATION:

MSc in Industrial Automation Engineering

National University of Colombia — Bogotá
2023 – 2025

BSc in Mechatronics Engineering

National University of Colombia — Bogotá
2017 – 2022

SKILLS:

- Machine Learning (scikit-learn, data pipelines)
- Backend Development (FastAPI)
- Edge Computing (Docker, ARM deployment)
- Time Series Analysis & Anomaly Detection
- Programming: Python, C++, Java
- Industrial Systems: PLC (Allen-Bradley), SCADA (Ignition), IIoT (ThingWorx)
- Embedded Systems: Arduino, ESP32, HCC2, Proton, PLCNext

PROJECTS

AI Dataset Reasoning Agent (LangGraph)

- Developed an AI agent for industrial datasets using LangGraph, focused on preprocessing, feature engineering, and baseline modeling for automated analysis.

CERTIFICATIONS:

2025	Professional Certificate - IBM Machine Learning Coursera – IBM
2025	Professional Certificate - IBM Data Science Coursera – IBM
2025	Specialized Course “IoT Systems and Industrial Applications with Design Thinking” Coursera – LearnQuest
2024	Specialized Course “ThingWorx Technical Essentials (Self-Paced)” PTC Field Academy
2023	Specialized Course “An Introduction to Programming the Internet of Things (IOT)” Coursera – University of California, Irvine
2020	Specialized Course “Mathematics for Machine Learning” Coursera – Imperial College London

LANGUAGES

- Spanish: Native
- English: B2 (technical communication)

HONORS:

- **National Recognition – Saber Pro Exam (2022):** Awarded by the Colombian Ministry of Education for achieving top performance in the national standardized exam for engineering graduates.
- **Best Undergraduate Thesis – Mechatronics Engineering (2022):** Awarded by the National University of Colombia for outstanding academic and technical contribution in undergraduate research.
- **Top Academic Performance – Mechatronics Engineering Cohort (2023):** Recognized by the National University of Colombia among the highest-performing graduates of the 2022-2 cohort.